

Real Business Cycles

Technology

$$Y_t = \theta_t F(K_t, A_t L_t)$$

where F displays standard neoclassical properties, and

$$A_t = (1 + g)^t A_0$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

where $A_0 > 0$, $g \geq 0$, $\theta > 0$, $\rho \in (0, 1)$, and $\{\varepsilon_{t+1}\}_{t=0}^{\infty}$ is a sequence of i.i.d random variables with distribution $\varepsilon_{t+1} \sim N(0, \sigma_{\varepsilon}^2)$. Notice that the evolution of $\ln \theta_{t+1}$ can be written as follows: $\ln(\frac{\theta_{t+1}}{\theta}) = \rho \ln(\frac{\theta_t}{\theta}) + \varepsilon_{t+1}$.

Capital evolves according to:

$$K_{t+1} = I_t + (1 - \delta)K_t$$

The resource constraint is:

$$C_t + I_t \leq Y_t$$

Population evolves according to:

$$N_t = (1 + n)^t N_0, \quad n \geq 0$$

The utility fn. of the representative household is:

$$\tilde{U} = \sum_{t=0}^{\infty} \tilde{\beta}^t u(\tilde{c}_t, 1 - \tilde{l}_t) \frac{N_t}{H}$$

where H is the fixed number of households (so $\frac{N_t}{H}$ gives the number of members of the household), $\tilde{c}_t = \frac{C_t}{N_t}$ is consumption for the representative member of each household (*i.e.*, consumption per capita), $1 - \tilde{l}_t$ is leisure per capita (total time is normalized to one and $\tilde{l}_t = \frac{L_t}{N_t}$ is the time spent working in per capita terms); $u(\cdot)$ is the per-period utility function, and $\tilde{\beta} \in (0, 1)$ is the discount factor.

For any variable X , define:

$$x_t \equiv \frac{X_t}{A_t N_t}$$

Recall that, in this model, $N_t \neq L_t$.

By CRS:

$$\begin{aligned} y_t &\equiv \frac{Y_t}{A_t N_t} = \frac{\theta_t F(K_t, A_t L_t)}{A_t N_t} \\ &= \theta_t F\left(\frac{K_t}{A_t N_t}, \frac{A_t L_t}{A_t N_t}\right) = \theta_t F\left(k_t, \frac{L_t}{N_t}\right) \\ &= \theta_t F(k_t, \tilde{l}_t) \end{aligned}$$

Remark. We can also write: $y_t = \theta_t F\left(\frac{K_t}{A_t N_t}, \frac{A_t L_t}{A_t N_t}\right) = \theta_t F(k_t, A_t l_t) = \theta_t F(k_t, A_0(1+g)^t l_t)$.

We can rewrite the resource constraint as follows:

$$C_t + K_{t+1} - (1 - \delta)K_t \leq Y_t$$

Dividing by $A_t N_t$:

$$\frac{C_t}{A_t N_t} + \frac{K_{t+1}}{A_{t+1} N_{t+1}} \frac{A_{t+1} N_{t+1}}{A_t N_t} - (1 - \delta) \frac{K_t}{A_t N_t} \leq \frac{Y_t}{A_t N_t}$$

Then:

$$c_t + k_{t+1}(1+g)(1+n) - (1-\delta)k_t \leq \theta_t F(k_t, \tilde{l}_t)$$

We can rewrite the utility function as follows:

$$\begin{aligned}
\tilde{U} &= \sum_{t=0}^{\infty} \tilde{\beta}^t u(\tilde{c}_t, 1 - \tilde{l}_t) \frac{N_t}{H} \\
&= \sum_{t=0}^{\infty} \tilde{\beta}^t u(A_t c_t, 1 - \tilde{l}_t) \frac{N_t}{H} \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \left(\frac{1+n}{1+\tilde{\rho}} \right)^t u(A_t c_t, 1 - \tilde{l}_t) \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t u(A_t c_t, 1 - \tilde{l}_t)
\end{aligned}$$

where $\beta \equiv \frac{1+n}{1+\tilde{\rho}} = \tilde{\beta}(1+n)$, with $\tilde{\rho} > n$, so $\beta \in (0, 1)$.

**Particular Case: Log Utility, Cobb-Douglas
Technology, 100% Depreciation**

Assume:

$$u(\tilde{c}_t, 1 - \tilde{l}_t) = \ln \tilde{c}_t + \gamma \ln(1 - \tilde{l}_t), \quad \gamma > 0$$

$$Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}, \quad \alpha \in (0, 1)$$

$$K_{t+1} = I_t \quad (\text{i.e., } \delta = 1)$$

Then:

$$\begin{aligned} y_t &= \theta_t F(k_t, \tilde{l}_t) \\ &= \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} \end{aligned}$$

The constraint becomes:

$$c_t + k_{t+1}(1 + g)(1 + n) \leq \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

The utility function becomes

$$\begin{aligned}
\tilde{U} &= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln(\tilde{c}_t) + \gamma \ln(1 - \tilde{l}_t) \right] = \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln(A_t c_t) + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln A_t + \ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \ln[A_0(1+g)^t] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t [\ln A_0 + t \ln(1+g)] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \left[\ln A_0 \sum_{t=0}^{\infty} \beta^t + \ln(1+g) \sum_{t=0}^{\infty} \beta^t t \right] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \frac{N_0}{H} \left[\ln A_0 \frac{1}{1-\beta} + \ln(1+g) \frac{\beta}{(1-\beta)^2} \right] + \frac{N_0}{H} \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right] \\
&= \text{constant}_1 + \text{positive constant}_2 \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]
\end{aligned}$$

Then we can work with:

$$U \equiv \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]$$

Planner's Problem

The planner chooses contingent plans for consumption (c_t), labor ($\tilde{l}_t$), and capital (k_{t+1}), in order to solve:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\ln c_t + \gamma \ln(1 - \tilde{l}_t) \right]$$

$$\text{s.t. } c_t + (1 + g)(1 + n)k_{t+1} \leq \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_0 > 0, \theta_0 > 0 \text{ given}$$

Remark 1. When choosing c_t , $\tilde{l}_t$ and k_{t+1} , the planner knows the value of θ_t (the past values too).

Remark 2. $\mathbb{E}_0$ denotes the conditional expectation as of time 0.

Remark 3. It's clear that the constraint will be binding at the optimum.

Remark 4. Our assumptions ensure that consumption and capital will be strictly positive at the optimum, and $\tilde{l}_t \in (0, 1)$.

Bellman Equation (BE)

For any period t :

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1})$$

$$\text{s.t.} \quad (1+g)(1+n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Strategy

First we'll use the FOC and the Envelope Condition to obtain some intuition. Then we'll solve the problem in closed form using the Guess-Verify method.

To facilitate the interpretation of the FOC we generalize notation and write $u(c_t)$ instead of $\ln c_t$ and $v(1 - \tilde{l}_t)$

instead of $\gamma \ln(1 - \tilde{l}_t)$. Assume $u' > 0$, $v' > 0$, $u'' < 0$ and $v'' < 0$. Also, use $\theta_t F(k_t, \tilde{l}_t)$ instead of $\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$. Finally, allow for a depreciation rate smaller than one. Then, BE becomes:

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} u(c_t) + v(1 - \tilde{l}_t) + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1})$$

$$\text{s.t.} \quad (1+g)(1+n)k_{t+1} = \theta_t F(k_t, \tilde{l}_t) + (1-\delta)k_t - c_t$$

$$\ln \theta_{t+1} = (1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

FOC (use the constraint to eliminate c_t and get the partial derivatives w.r.t. k_{t+1} and $\tilde{l}_t$).

$$(1) \quad u'(c_t)(1+g)(1+n) = \beta \mathbb{E}_t V_1(k_{t+1}, \theta_{t+1})$$

$$(2) \quad v'(1 - \tilde{l}_t) = u'(c_t) \theta_t F_2(k_t, \tilde{l}_t)$$

$$(3) \quad c_t = \theta_t F(k_t, \tilde{l}_t) + (1-\delta)k_t - (1+g)(1+n)k_{t+1}$$

Envelope

$$(4) \quad V_1(k_t, \theta_t) = u'(c_t) [\theta_t F_1(k_t, \tilde{l}_t) + (1 - \delta)]$$

Shifting (4) one period ahead and substituting into (1):

$$(5) \quad u'(c_t)(1+g)(1+n) = \beta \mathbb{E}_t \left\{ u'(c_{t+1}) [\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)] \right\}$$

Summarizing:

$$(6) \quad v'(1 - \tilde{l}_t) = u'(c_t) \theta_t F_2(k_t, \tilde{l}_t)$$

$$(7) \quad u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1+g)(1+n)} \right\}$$

$$(8) \quad c_t + (1 + g)(1 + n)k_{t+1} = \theta_t F(k_t, \tilde{l}_t) + (1 - \delta)k_t$$

Hence, we've obtained an intratemporal condition between consumption and leisure, (6), an intertemporal condition for consumption, (7), and the budget constraint, (8). We have three equations (per period) to solve for the evolution of three variables: c_t , $\tilde{l}_t$, and k_{t+1} .

Combining expressions (6) and (7) we can obtain three additional conditions (which are redundant from the point of view of the model's solution).

Substituting (6) into both sides of (7) we get an intertemporal condition for labor/leisure:

$$(9) \quad v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \begin{aligned} & \beta v'(1 - \tilde{l}_{t+1}) \times \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \\ & \times \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \end{aligned} \right\}$$

Substituting (6) into the left-hand side of (7) we get an intertemporal condition between l_t and c_{t+1} :

$$(10) \quad v'(1 - \tilde{l}_t) = \beta \mathbb{E}_t \left\{ u'(c_{t+1}) \left[\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta) \right] \theta_t F_2(k_t, \tilde{l}_t) \right\}$$

Finally, substituting (6) into the right-hand side of (7) we get an intertemporal condition between c_t and l_{t+1} :

$$(11) \quad u'(c_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n) \theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \right\}$$

Interpretation of (6).

Consider a small increase in current labor, $d\tilde{l}_t = \Delta > 0$. Hence, current leisure goes down by $d(1 - \tilde{l}_t) = -\Delta$. This induces a change in utility equal to: $v'(1 - \tilde{l}_t)d(1 - \tilde{l}_t) = -v'(1 - \tilde{l}_t)\Delta < 0$.

The extra amount of labor increases current output by: $dy_t = \theta_t F_2(k_t, \tilde{l}_t)d\tilde{l}_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$ (k_t is given). Imagine you use all this extra output to increase current consumption. Then: $dc_t = dy_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$. This increases utility by $u'(c_t)dc_t = u'(c_t)\theta_t F_2(k_t, \tilde{l}_t)\Delta$.

Equation (6) says that, at the optimum, the reduction in utility induced by a reduction in leisure (increase in labor) must be exactly compensated by the increase in utility induced by the extra consumption (obtained through the increase in labor). [Notice that Δ is in both terms, so it cancels out]

Another way: from (6) we get: $\frac{v'(1-\tilde{l}_t)}{u'(c_t)} = \theta_t F_2(k_t, \tilde{l}_t)$. This says that, at the optimum, the marginal rate of substitution between leisure and consumption, $\frac{v'(1-\tilde{l}_t)}{u'(c_t)}$, must be equal to the marginal rate of transformation between them, which equals the marginal product of labor, $\theta_t F_2(k_t, \tilde{l}_t)$.

Interpretation of (7) [Euler Equation for Consumption]

Consider a small reduction in cons. at t , $dc_t = -\Delta < 0$. This changes utility by $u'(c_t)dc_t = -u'(c_t)\Delta < 0$.

Now use the reduction in consumption to increase the capital stock. From the resource constraint we get (keeping output constant): $dc_t + (1+g)(1+n)dk_{t+1} = 0$. Then: $dk_{t+1} = \frac{\Delta}{(1+g)(1+n)}$.

The increase in capital increases next-period output by: $dy_{t+1} = \theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1} = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})\Delta}{(1+g)(1+n)}$. In addition, you get the undepreciated capital: $(1-\delta)dk_{t+1} = \frac{(1-\delta)\Delta}{(1+g)(1+n)}$. Imagine you use all these extra resources to increase next-period consumption. Then: $dc_{t+1} = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}\Delta$. The increase in expected utility (recall θ_{t+1} is unknown at t) induced by the extra consumption is:

$$\mathbb{E}_t\{\beta u'(c_{t+1})dc_{t+1}\} = \mathbb{E}_t\left\{\beta u'(c_{t+1})\frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}\right\}\Delta.$$

Hence, (7) says that, at the optimum, the reduction in utility induced by a reduction in current consumption must be exactly compensated by the increase in expected utility induced by next period's additional consumption.

Think about the case with no uncertainty. Then, (8) becomes: $u'(c_t) = \beta u'(c_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$. We can rewrite this as follows: $\frac{u'(c_t)}{\beta u'(c_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$. This says that, at the optimum, the marginal rate of substitution between current and next-period consumption, $\frac{u'(c_t)}{\beta u'(c_{t+1})}$, must equal the marginal rate of transformation between current and next-period consumption, $\frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$.

Interpretation of (9) [Euler Equation for Labor/Leisure]

Consider a small reduction in current leisure, $d(1 - \tilde{l}_t) = -\Delta$. This changes utility by $v'(1 - \tilde{l}_t)d(1 - \tilde{l}_t) = -v'(1 - \tilde{l}_t)\Delta < 0$.

The reduction in leisure increases labor by: $d\tilde{l}_t = \Delta$. This increases current output by: $dy_t = \theta_t F_2(k_t, \tilde{l}_t)d\tilde{l}_t = \theta_t F_2(k_t, \tilde{l}_t)\Delta$.

Imagine you use this extra output to increase the capital stock. From the resource constraint we get (keeping c_t constant): $(1 + g)(1 + n)dk_{t+1} = dy_t$. Then: $dk_{t+1} = \frac{\theta_t F_2(k_t, \tilde{l}_t)\Delta}{(1+g)(1+n)}$.

For a given level of next-period labor, the increase in capital would increase next-period output by: $dy_{t+1} = \theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1}$.

In addition, you would get the undepreciated capital: $(1 - \delta)dk_{t+1}$.

Hence, total additional resources would be $dy_{t+1} + (1 - \delta)dk_{t+1} = [\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)] dk_{t+1}$.

Imagine you use these extra resources to reduce the amount of next-period work. How much would you be able to reduce $\tilde{l}_{t+1}$? From the prod. fn. we know that $dy_{t+1} = \theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1})dk_{t+1} + \theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})d\tilde{l}_{t+1}$. Then $dy_{t+1} + (1-\delta)dk_{t+1} = \left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] dk_{t+1} + \theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})d\tilde{l}_{t+1}$.

Setting $dy_{t+1} + (1-\delta)dk_{t+1} = 0$ we get:

$$\begin{aligned} d\tilde{l}_{t+1} &= -\frac{\left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] dk_{t+1}}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})} \\ &= -\frac{\left[\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta) \right] \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{(1+g)(1+n)}}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})} \\ &= -\frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})}. \end{aligned}$$

Then, next-period leisure increases by:

$$d(1 - \tilde{l}_{t+1}) = \frac{\theta_{t+1}F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t) \Delta}{\theta_{t+1}F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

This increases exp. ut. by: $\mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) d(1 - \tilde{l}_{t+1}) \right\} = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})} \right\} \Delta$.

Hence, (9) says that, at the optimum, the reduction in utility induced by a reduction in current leisure must be exactly compensated by the increase in expected utility induced by next period's additional leisure.

Think about the case with no uncertainty. Then: $v'(1 - \tilde{l}_t) = \beta v'(1 - \tilde{l}_{t+1}) \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}$.

We can rewrite this as follows:

$$\frac{v'(1 - \tilde{l}_t)}{\beta v'(1 - \tilde{l}_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1 - \delta)}{(1 + g)(1 + n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

This says that, at the optimum, the marginal rate of substitution between current and next-period leisure must equal the marginal rate of transformation between current and next-period leisure.

The interpretation of (10) and (11) can be done analogously.

Wealth and Substitution Effects

Consider the deterministic case, for intuition.

Wealth Effect

An increase in θ_t has a positive wealth effect, since $y_t = \theta_t F(k_t, \tilde{l}_t)$ goes up for given values of k_t and $\tilde{l}_t$.

If consumption and leisure are normal goods, then c_t and $1 - \tilde{l}_t$ will go up (and $\tilde{l}_t$ will go down) for every t .

If the shock is temporary (i.e., θ_t increases but $\theta_{t+1}, \theta_{t+2}, \dots$ remain constant) then c_t will increase only a little, $\tilde{l}_t$ will go down only a little, and k_{t+1} will go up a lot in order to increase future output possibilities and then future consumption and leisure.

If the shock is permanent (i.e., θ_t increases for all t) the increase in c_t and the reduction in $\tilde{l}_t$ will be big, and the effect on k_{t+1} will be basically zero, since there's no incentive to shift resources towards the future.

Substitution Effects

The analysis of the wealth effects did not consider the impact of θ_t on marginal products.

An increase of θ_t increases the current marginal product of labor. This has an intratemporal substitution effect: reduce current leisure (increase labor) and increase current consumption. This can be seen from the intratemporal FOC: $\frac{v'(1-\tilde{l}_t)}{u'(c_t)} = \theta_t F_2(k_t, \tilde{l}_t)$.

Remark. Strictly speaking, the expression above does not imply that consumption and labor have to go up after the positive technology shock. However, to think about the substitution effect we need to eliminate the wealth effect. If there's no wealth effect, consumption and leisure have to go in opposite directions (with no extra wealth, to consume more you have to work more). The same applies to the comments below.

An increase of θ_t relative to θ_{t+1} increases the current marginal product of labor relative to tomorrow's. For a given marginal product of capital, this has an intertemporal substitution effect on labor/leisure: work more in t , work less in $t + 1$. From (9):

$$\frac{v'(1-\tilde{l}_t)}{\beta v'(1-\tilde{l}_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t, \tilde{l}_t)}{\theta_{t+1} F_2(k_{t+1}, \tilde{l}_{t+1})}.$$

The latter effect requires an increase in investment in t , and then k_{t+1} increases.

In period t , an expected increase in θ_{t+1} increases the expected marginal product of capital. This has an intertemporal substitution effect on consumption: reduce c_t and increase c_{t+1} . From (7): $\frac{u'(c_t)}{\beta u'(c_{t+1})} = \frac{\theta_{t+1} F_1(k_{t+1}, \tilde{l}_{t+1}) + (1-\delta)}{(1+g)(1+n)}$.

In period t , an expected increase in θ_{t+1} increases the expected marginal product of capital. For given relative marginal products of labor, this has an intert. subst. effect on labor: increase $\tilde{l}_t$ and reduce $\tilde{l}_{t+1}$. From (9):

$$\frac{v'(1-\tilde{l}_t)}{\beta v'(1-\tilde{l}_{t+1})} = \frac{\theta_{t+1}F_1(k_{t+1},\tilde{l}_{t+1})+(1-\delta)}{(1+g)(1+n)} \frac{\theta_t F_2(k_t,\tilde{l}_t)}{\theta_{t+1}F_2(k_{t+1},\tilde{l}_{t+1})}.$$

The two effects above imply that the increase in the expected marginal product of capital increases investment in period t , and hence increases k_{t+1} .

Define:

$$R_t \equiv \theta_t F_1(k_t, \tilde{l}_t) \quad MPK_t$$

$$w_t \equiv \theta_t F_2(k_t, \tilde{l}_t) A_t \quad MPL_t$$

Remark: $MPK_t = \frac{\partial Y_t}{\partial K_t} = \theta_t F_1(K_t, A_t L_t) = \theta_t F_1(k_t, \tilde{l}_t)$, where the last equality follows from the homogeneity of degree zero of F_1 in K and L (dividing by $A_t N_t$). Analogously, $MPL_t = \frac{\partial Y_t}{\partial L_t} = \theta_t \frac{\partial F(K_t, A_t L_t)}{\partial L_t} = \theta_t F_2(K_t, A_t L_t) A_t = \theta_t F_2(k_t, \tilde{l}_t) A_t$, where the last equality follows from the homogeneity of degree zero of F_2 in K and L (dividing by $A_t N_t$).

Since F is homogeneous of degree one in K and L , Euler's Theorem implies $Y_t = \theta_t \frac{\partial F}{\partial L_t} L_t + \theta_t \frac{\partial F}{\partial K_t} K_t \Rightarrow Y_t = \theta_t F_2 A_t L_t + \theta_t F_1 K_t \Rightarrow Y_t = w_t L_t + R_t K_t$. Dividing by $A_t N_t$:

$$y_t = w_t l_t + R_t k_t$$

$$= \frac{w_t}{A_t} \tilde{l}_t + R_t k_t$$

Using the expressions above, we can rewrite the FOCs as follows:

$$v'(1 - \tilde{l}_t) = u'(c_t) \frac{w_t}{A_t}$$

$$u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \right\}$$

$$c_t + (1 + g)(1 + n)k_{t+1} = \frac{w_t}{A_t} \tilde{l}_t + R_t k_t + (1 - \delta)k_t$$

The Euler equation for labor/leisure becomes:

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \frac{\frac{w_t}{A_t}}{\frac{w_{t+1}}{A_{t+1}}} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{(1 + g)(1 + n)} \frac{w_t}{w_{t+1}} \frac{A_{t+1}}{A_t} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{R_{t+1} + (1 - \delta)}{1 + n} \frac{w_t}{w_{t+1}} \right\}$$

where the last line uses $\frac{A_{t+1}}{A_t} = 1 + g$.

Notice that the expressions above are exactly the ones we would get in the decentralized version of the model. In this case, w_t would be the wage rate at t , R_t would be the rental rate of capital at t , and $y_t = w_t l_t + R_t k_t$ is the zero-profit condition. We can also reinterpret the wealth and substitution effects (e.g., the wealth and substitution effects of a temporary change in wages).

If we add a credit market to the decentralized version, the Euler equations for consumption and labor can be written as follows:

$$u'(c_t) = \mathbb{E}_t \left\{ \beta u'(c_{t+1}) \frac{1+r_t}{(1+g)(1+n)} \right\}$$

$$v'(1 - \tilde{l}_t) = \mathbb{E}_t \left\{ \beta v'(1 - \tilde{l}_{t+1}) \frac{1+r_t}{1+n} \frac{w_t}{w_{t+1}} \right\}$$

where r_t is the bond's real interest rate (determined in period t , paid in $t+1$). In this case, $1+r_t$ can be moved outside the expectations operator (because r_t is known in period t).

A Closed-Form Solution for our Special Case

The Bellman Equation is:

$$V(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) \\ + \beta \mathbb{E}_t V(k_{t+1}, \theta_{t+1}) \end{array} \right\}$$

$$\text{s.t.} \quad (1 + g)(1 + n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Guess:

$$V^G(k_t, \theta_t) = D + E \ln k_t + F \ln \theta_t$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t, k_{t+1}} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) \\ + \beta \mathbb{E}_t \{ D + E \ln k_{t+1} + F \ln \theta_{t+1} \} \end{array} \right\}$$

$$\text{s.t.} \quad (1 + g)(1 + n)k_{t+1} = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t$$

$$\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$$

$$k_t, \theta_t \text{ given}$$

Then

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \\ \beta \mathbb{E}_t \left\{ \begin{array}{l} D + E \ln \left(\frac{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}{(1+g)(1+n)} \right) \\ + F[(1-\rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}] \end{array} \right\} \end{array} \right\}$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta D \\ + \beta E \ln \left(\frac{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}{(1+g)(1+n)} \right) \\ + \beta F(1 - \rho) \ln \theta \\ + \beta F \rho \ln \theta_t + \beta F \mathbb{E}_t \varepsilon_{t+1} \end{array} \right\}$$

Then:

$$\tilde{V}(k_t, \theta_t) \equiv \max_{c_t, \tilde{l}_t} \left\{ \begin{array}{l} \ln c_t + \gamma \ln(1 - \tilde{l}_t) + \beta D \\ + \beta E \ln \left(\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t \right) \\ - \beta E \ln[(1+g)(1+n)] \\ + \beta F(1 - \rho) \ln \theta + \beta F \rho \ln \theta_t \end{array} \right\}$$

FOC:

$$c_t : \quad \frac{1}{c_t} = \beta E \frac{1}{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}$$

$$\tilde{l}_t : \quad \frac{\gamma}{1-\tilde{l}_t} = \beta E \frac{(1-\alpha)\theta_t k_t^\alpha \tilde{l}_t^{-\alpha}}{\theta_t k_t^\alpha \tilde{l}_t^{1-\alpha} - c_t}$$

From the FOC for c_t we get:

$$c_t = \frac{1}{1 + \beta E} \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$$

Substituting the expression above into the FOC for $\tilde{l}_t$ we get:

$$\tilde{l}_t = \frac{(1 - \alpha)(1 + \beta E)}{(1 - \alpha)(1 + \beta E) + \gamma} \equiv \tilde{l}(E)$$

Remark. $\tilde{l}$ is constant over time.

Substituting $\tilde{l}_t = \tilde{l}(E)$ into the expression for c_t :

$$c_t = \frac{1}{1 + \beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha}$$

Substituting the previous expressions into the resource constraint:

$$k_{t+1} = \frac{1}{(1 + g)(1 + n)} \frac{\beta E}{1 + \beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha}$$

Substituting into the objective function:

$$\begin{aligned}\tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\ln \left(\frac{1}{1+\beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha} \right) + \gamma \ln(1 - \tilde{l}(E)) \\ &+ \beta D + \beta E \ln \left(\frac{1}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} \theta_t k_t^\alpha [\tilde{l}(E)]^{1-\alpha} \right) \\ &+ \beta F[(1-\rho) \ln \theta + \rho \ln \theta_t] \end{aligned} \right\} \\ \tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\ln \left(\frac{1}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) + \ln \theta_t + \alpha \ln k_t + \gamma \ln(1 - \tilde{l}(E)) \\ &+ \beta D + \beta E \ln \left(\frac{1}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) \\ &+ \beta E \ln \theta_t + \beta E \alpha \ln k_t + \beta F(1-\rho) \ln \theta + \beta F \rho \ln \theta_t \end{aligned} \right\} \\ \tilde{V}(k_t, \theta_t) &= \left\{ \begin{aligned} &\left[\ln \left(\frac{1}{1+\beta E} [\tilde{l}(E)]^{1-\alpha} \right) + \gamma \ln(1 - \tilde{l}(E)) + \beta D \right. \\ &\quad \left. + \beta E \ln \left(\frac{[\tilde{l}(E)]^{1-\alpha}}{(1+g)(1+n)} \frac{\beta E}{1+\beta E} \right) + \beta F(1-\rho) \ln \theta \right] \\ &+ (1 + \beta E + \rho \beta F) \ln \theta_t + \alpha(1 + \beta E) \ln k_t \end{aligned} \right\}\end{aligned}$$

Then:

$$\tilde{V}(k_t, \theta_t) = \Omega(D, E, F) + \alpha(1 + \beta E) \ln k_t + (1 + \beta E + \rho\beta F) \ln \theta_t$$

For $\tilde{V}(k_t, \theta_t)$ to equal $V^G(k_t, \theta_t) \forall (k_t, \theta_t)$ we need

$$\begin{aligned} D &= \Omega(D, E, F) \\ E &= \alpha(1 + \beta E) \\ F &= 1 + \beta E + \rho\beta F \end{aligned}$$

From the second equation we obtain:

$$E = \frac{\alpha}{1 - \alpha\beta}$$

Substituting into the third equation:

$$F = \frac{1}{(1 - \alpha\beta)(1 - \rho\beta)}$$

Substituting the values of E and F into the first equation we can obtain the value of the additive constant, D .

Using the results above we conclude:

$$V(k_t, \theta_t) = D + \frac{\alpha}{1 - \alpha\beta} \ln k_t + \frac{1}{(1 - \alpha\beta)(1 - \rho\beta)} \ln \theta_t$$

and

$$\tilde{l} = \frac{1 - \alpha}{(1 - \alpha) + \gamma(1 - \alpha\beta)} \in (0, 1)$$

$$c_t = (1 - \alpha\beta) \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$$

$$k_{t+1} = \frac{\alpha\beta}{(1 + g)(1 + n)} \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$$

Remark 1. Notice that $c_t = (1 - \alpha\beta) y_t$ and $k_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} y_t$. Hence, constant saving rate, $s = \alpha\beta$.

Remark 2. Labor and leisure are constant. Therefore, wealth and substitution effects cancel each other.

Nonstochastic Steady State (Balanced Growth)

Since $\tilde{l}$ is constant, it's clear that:

$$\tilde{l}_{nss} = \tilde{l} = \frac{1 - \alpha}{(1 - \alpha) + \gamma(1 - \alpha\beta)}$$

Recall $\ln \theta_{t+1} = (1 - \rho) \ln \theta + \rho \ln \theta_t + \varepsilon_{t+1}$ with $\varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$ and $\rho \in (0, 1)$. Setting $\varepsilon_{t+1} = \mathbb{E}\varepsilon_{t+1} = 0$ and $\theta_{t+1} = \theta_t = \theta_{nss}$ we get:

$$\theta_{nss} = \theta$$

Setting $\theta_t = \theta$ and $k_{t+1} = k_t = k_{nss}$ in $k_{t+1} = \frac{\alpha\beta}{(1+g)(1+n)} \theta_t k_t^\alpha \tilde{l}^{1-\alpha}$ we get: $k_{nss} = \frac{\alpha\beta}{(1+g)(1+n)} \theta k_{nss}^\alpha \tilde{l}^{1-\alpha}$.
Then:

$$k_{nss} = \left[\frac{\alpha\beta\theta}{(1+g)(1+n)} \right]^{\frac{1}{1-\alpha}} \tilde{l}$$

From $y_t = \theta_t k_t^\alpha \tilde{l}_t^{1-\alpha}$ we get:

$$y_{nss} = \theta k_{nss}^\alpha \tilde{l}_{nss}^{1-\alpha}$$

From $c_t = (1 - \alpha\beta) y_t$ we get:

$$c_{nss} = (1 - \alpha\beta) y_{nss}$$

Recall that: $x_t \equiv \frac{X_t}{A_t N_t} = \frac{\tilde{x}_t}{A_t}$. Therefore, if x is constant in the nonstochastic steady state, the aggregate variable X grows at (gross) rate $(1 + g)(1 + n)$, and the per-capita variable $\tilde{x}_t$ grows at (gross) rate $1 + g$.

Calibration

Strategy: Choose parameter values so that the balanced-growth path of the model economy matches certain long-term features of the measured economy.

Remark. Go back to the case with $\delta < 1$ (because we want to calibrate its value).

Calibration of n

The rate of population growth, n , can be measured directly, as the average population growth over a long period.

Calibration of g

In the nonstochastic steady state of the model, g equals the growth rate of per capita GDP. Hence, g can be measured directly as the average growth of per capita GDP over a long period.

Remark. Later we'll provide another way of calibrating g .

Calibration of α

From $Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}$ we get: $\frac{\partial Y_t}{\partial L_t} = (1 - \alpha) \frac{Y_t}{L_t}$.

Now think about the behavior of firms in the decentralized version of the model.

Real profits are: $Y_t - w_t L_t - R_t K_t$.

Profit maximization with respect to L_t : $\frac{\partial Y_t}{\partial L_t} = w_t$.

Then: $(1 - \alpha) \frac{Y_t}{L_t} = w_t$.

Then: $1 - \alpha = \frac{w_t L_t}{Y_t}$.

Therefore, $1 - \alpha$ can be measured as the average share of labor income over a long period.

Calibration of ρ and σ_ε^2

The production function is: $Y_t = \theta_t K_t^\alpha (A_t L_t)^{1-\alpha}$

Then: $Y_t = (\theta_t A_t^{1-\alpha}) K_t^\alpha L_t^{1-\alpha} = z_t K_t^\alpha L_t^{1-\alpha}$.

Then: $\ln Y_t = \ln z_t + \alpha \ln K_t + (1 - \alpha) \ln L_t$.

Then: $\ln z_t = \ln Y_t - \alpha \ln K_t - (1 - \alpha) \ln L_t$.

Given our value of α , the r-h-s can be estimated from data. This gives an estimate of $\ln z_t$ (Solow Residual).

We know $\ln z_t = \ln(\theta_t A_t^{1-\alpha})$.

Then: $\ln z_t = (1 - \alpha) \ln A_t + \ln \theta_t$.

Recall $A_t = (1 + g)^t A_0$.

Then: $\ln A_t = \ln A_0 + \ln(1 + g) \times t$.

Then: $\ln z_t = (1 - \alpha)[\ln A_0 + \ln(1 + g) \times t] + \ln \theta_t$.

Then: $\ln z_t = (1 - \alpha) \ln A_0 + (1 - \alpha) \ln(1 + g) \times t + \ln \theta_t$.

Then: $\ln z_t = \beta_0 + \beta_1 t + u_t$.

Then, fitting a linear trend to $\ln z_t$, and given our calibrated value for α , we can estimate:

(i) A_0 from $\beta_0 = (1 - \alpha) \ln A_0$ (this is not very interesting for our purposes);

(ii) g , from $\beta_1 = (1 - \alpha) \ln(1 + g)$;

(iii) the noise $u_t = \ln \theta_t$ for each t (from the regression residuals)

With the estimated time series for $\ln \theta_t$ we can run a regression of the form

$$\ln \theta_t = \vartheta + \rho \ln \theta_{t-1} + \varepsilon_t.$$

This gives:

- (i) an estimate of ρ ;
- (ii) an estimate of θ , from $\vartheta = (1 - \rho) \ln \theta$ (again, not very interesting for our purposes).
- (iii) an estimate of ε_t for each t (from the regression residuals).

With these residuals we can calibrate σ_ε^2 .

Calibration of δ

We know: $K_{t+1} = I_t + (1 - \delta)K_t$.

Then: $\frac{K_{t+1}}{A_{t+1}N_{t+1}} \frac{A_{t+1}N_{t+1}}{A_t N_t} = \frac{I_t}{A_t N_t} + (1 - \delta) \frac{K_t}{A_t N_t}$.

Then: $k_{t+1}(1 + g)(1 + n) = \frac{I_t}{K_t} + (1 - \delta)k_t$.

In the nonstochastic steady-state:

$$k_{nss}(1 + g)(1 + n) = \frac{I_{nss}}{K_{nss}} + (1 - \delta)k_{nss}.$$

Then:

$$\delta = \frac{I_{nss}}{K_{nss}} + 1 - (1 + g)(1 + n).$$

Hence, with a long time series of the investment/capital ratio $\frac{I_t}{K_t}$ ($= \frac{I_t}{k_t}$) we can measure the depreciation rate (recall we have already calibrated g and n).

Calibration of β

Evaluating the Euler Equation for consumption at the nss:

$$u'(c_{nss}) = \beta u'(c_{nss}) \frac{\alpha \theta k_{nss}^{\alpha-1} \tilde{l}^{1-\alpha} + (1-\delta)}{(1+g)(1+n)}.$$

$$\text{Then: } (1+g)(1+n) = \beta \left[\alpha \frac{\theta k_{nss}^{\alpha} \tilde{l}^{1-\alpha}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g)(1+n) = \beta \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g)(1+n) = \tilde{\beta}(1+n) \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } (1+g) = \tilde{\beta} \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right].$$

$$\text{Then: } \tilde{\beta} = (1+g) \left[\alpha \frac{y_{nss}}{k_{nss}} + 1 - \delta \right]^{-1}.$$

Therefore, with a long time series of the output/capital ratio $\frac{Y_t}{K_t}$ ($= \frac{y_t}{k_t}$) we can measure the discount factor $\tilde{\beta}$ (recall we have already calibrated α , δ , and g).

Calibration of γ

From the intratemporal FOC:

$$v'(1 - \tilde{l}_{nss}) = u'(c_{nss})\theta F_2(k_{nss}, \tilde{l}_{nss}).$$

For logarithmic preferences and Cobb-Douglas technology we get:

$$\frac{\gamma}{1 - \tilde{l}_{nss}} = \frac{1}{c_{nss}}(1 - \alpha)\theta k_{nss}^{\alpha} \tilde{l}_{nss}^{-\alpha}.$$

$$\text{Then: } \frac{\gamma}{1 - \tilde{l}_{nss}} = \frac{1}{c_{nss}}(1 - \alpha)\frac{y_{nss}}{\tilde{l}_{nss}}.$$

$$\text{Then: } \gamma = (1 - \alpha)\frac{y_{nss}}{c_{nss}}\frac{1 - \tilde{l}_{nss}}{\tilde{l}_{nss}}.$$

Therefore, given an estimate of the long-run fraction of time devoted to market activities $\tilde{l}_{nss}$, and a long time series of the output/consumption ratio $\frac{Y_t}{C_t}$ ($= \frac{y_t}{c_t}$), we can measure the preference parameter γ (recall we have already calibrated α).

Example

For quarterly US data

α	δ	ρ	σ_ε	g	$\tilde{\beta}$	γ	n
0.4	0.012	0.95	0.007	0.0156/4	0.987	1.78	0.012/4

Notice that: $\beta = \tilde{\beta}(1 + n) = 0.9988 < 1$.

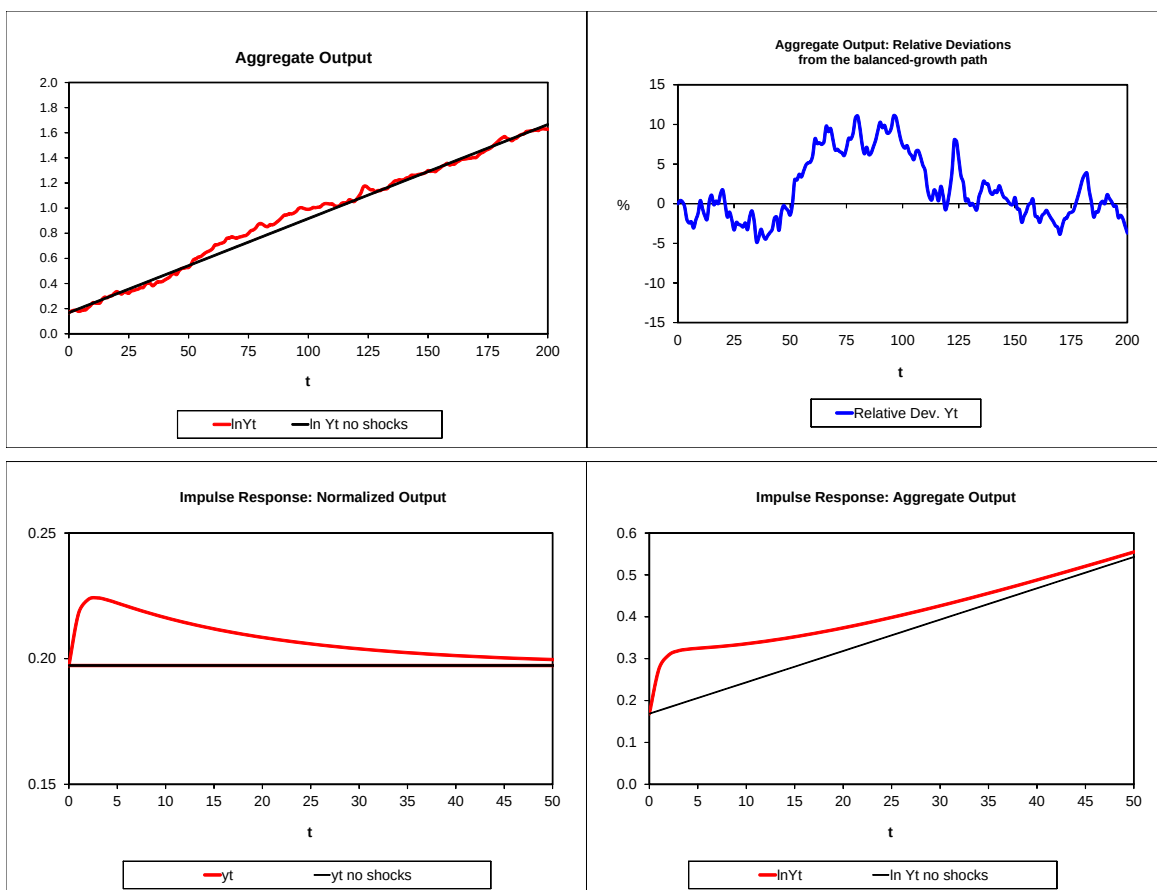
Adapted from

Cooley, T. and Prescott, E., "Economic Growth and Business Cycles", Chapter 1 in *Frontiers of Business Cycle Research*, edited by Thomas Cooley, Princeton University Press, 1995.

Plots ($\delta = 1$, parameters not calibrated, just an example)

Start at the nonstochastic steady state (no transition).

Set $A_0 = 6$, $L_0 = 1$, $\theta = 1$, $\alpha = 0.3$, $\tilde{\beta} = 0.99$, $n = 0.0025$, $g = 0.005$, $\gamma = 2$, $\rho = 0.95$, $\sigma_\varepsilon = 0.01$.
Then: $\tilde{l} = 0.33$, $k_{nss} = 0.058$, $y_{nss} = 0.197$.



Hodrick-Prescott (HP) Filter

Let $\{y_t\}$ be the time series to be decomposed into the trend component, $\{\tau_t\}$, and the cyclical component $\{c_t\}$, in such a way that $y_t = \tau_t + c_t$ (e.g., y_t may be the natural log of GDP).

The HP filter chooses the $\{\tau_t\}$ that solves:

$$\min \sum_{t=1}^T (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2$$

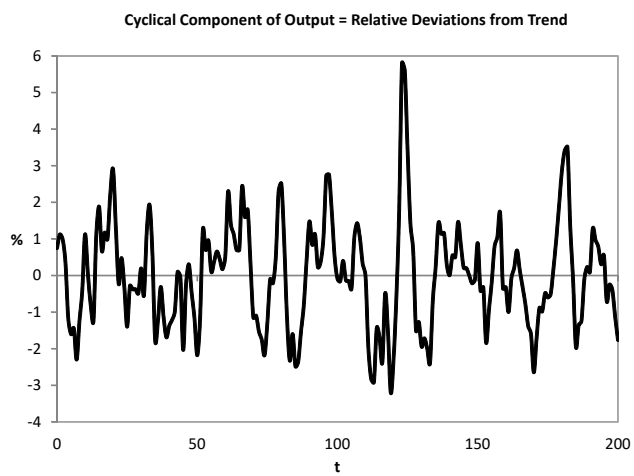
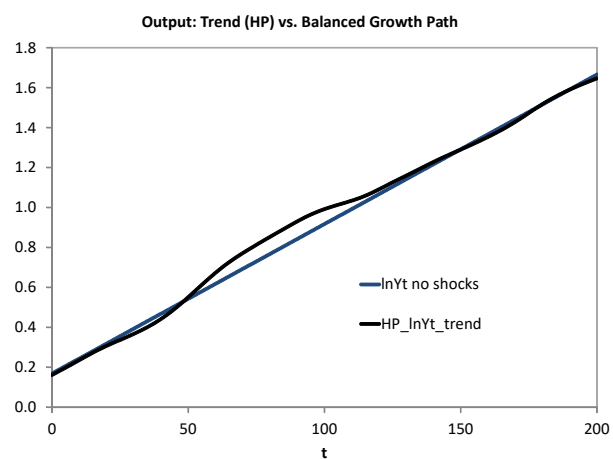
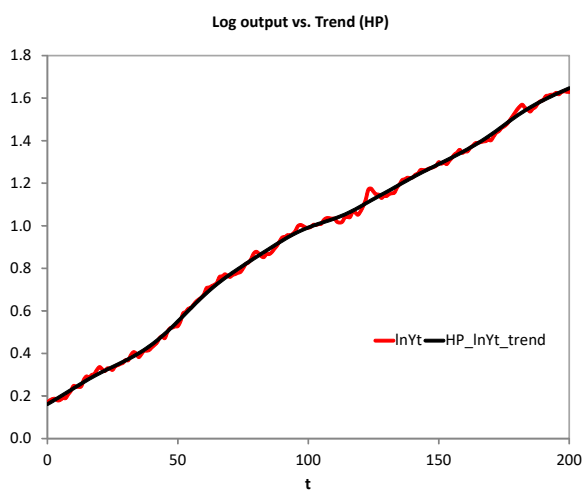
where λ is the penalty attached to the volatility of the trend.

We have: $\lambda = 0 \Rightarrow \tau_t = y_t$, and $\lambda \rightarrow \infty \Rightarrow \tau_t = \tau_0 + at$ (i.e., linear trend)

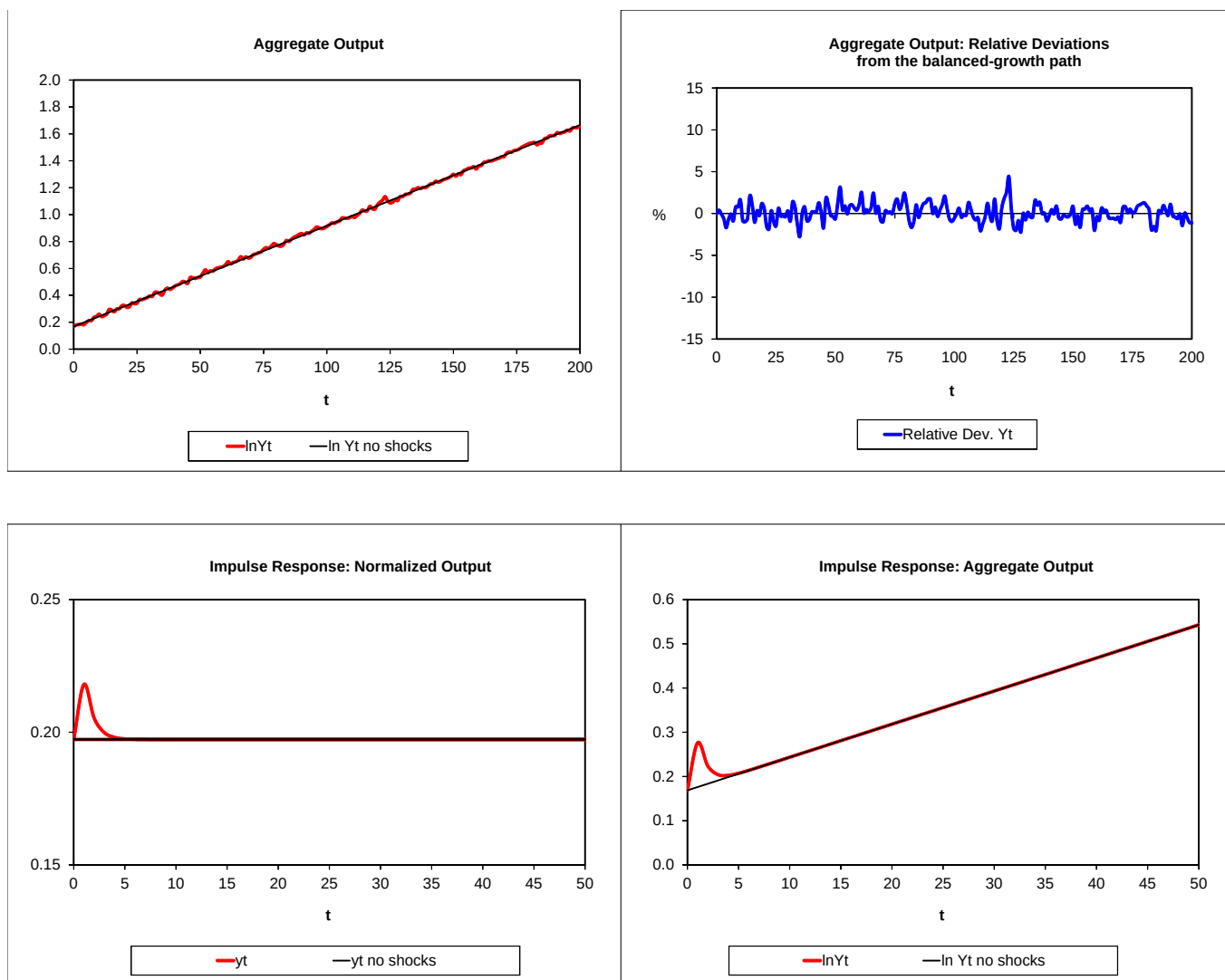
The standard value for quarterly data is $\lambda = 1600$. For annual data: $\lambda = 100$ (Ravn and Uhlig suggest $\lambda = 6.25$). For monthly data: $\lambda = 14400$ (Ravn and Uhlig suggest $\lambda = 129600$).

Remark. The cyclical component obtained from the HP filter is invariant to the presence of a linear trend.

Use the HP filter to detrend the series.



Reduce ρ to 0.1



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